SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

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Week 12 Progress Report

Details

- **Name of Student:** Tirthak Girish Likhar
- **PRN:** 22070521041
- **Name of Industry:** SS Infotech Pvt. Ltd.
- **Reporting Week: (23/03/2026 to 27/03/2026)**
- **Institute Mentor Name:** Dr. Shreyas Hole
- **Industry Mentor Name:** Mr. Viraj Patle

Overview of the Week

- Week 12 focused on the fourth attack simulation of the lab: an SSH Brute Force attack using Hydra v9.5. The simulation was run from the Ubuntu 22.04 attacker VM (192.168.1.7) targeting the SSH service running on the Wazuh Server (192.168.1.6). A custom password wordlist was created manually to control the attack and ensure reliable detection.

- Hydra attempted 9 passwords against the SSH service. Eight attempts failed with incorrect passwords, and one correct password was found in approximately 7 seconds. Throughout the attack, Wazuh was monitoring /var/log/auth.log on the server and detected all login attempts in real time.

- Wazuh successfully detected the attack. Multiple failed SSH login alerts were generated under Rule 5760 at Level 5 (one per failed attempt). When Wazuh correlated 8 or more failed attempts from the same source IP (192.168.1.7), it automatically escalated the alert to Rule 5763 at Level 10 — classifying it as a Brute Force attack under MITRE ATT&CK Technique T1110 (Brute Force), Credential Access tactic. A successful authentication alert also fired under Rule 5715 when Hydra found the correct password.

- Additionally, Medusa v2.2 was installed on the attacker VM as an alternative brute force tool for future use. The Week 12 simulation completes the four-attack demonstration set for the internship project — covering failed login (T1078), EICAR test file detection, FIM file creation trigger (Rule 554), and SSH brute force (T1110).

Objectives of the Week

- To perform Attack Simulation 4: SSH Brute Force attack using Hydra v9.5 from the Ubuntu 22.04 attacker VM (192.168.1.7) against the Wazuh Server SSH service (192.168.1.6).
- To create a custom password wordlist and use it with Hydra to simulate a realistic brute force credential attack.
- To verify that Wazuh detects individual failed SSH login attempts and generates Rule 5760 (Level 5) alerts for each failure.

- To confirm that Wazuh correlates multiple failed attempts into a single Rule 5763 (Level 10) Brute Force alert and correctly tags it with MITRE T1110 and the Credential Access tactic.
- To verify that the successful authentication alert (Rule 5715) fires when the correct password is discovered by Hydra.
- To confirm that the source attacker IP (192.168.1.7) is captured in all alert metadata.
- To verify compliance framework tags triggered by the attack: GDPR, HIPAA, PCI-DSS, and NIST 800-53.
- To install Medusa v2.2 on the attacker VM as an alternative brute force tool for future attack simulations.

Key Accomplishments

- Performed SSH Brute Force Attack Simulation (Attack Simulation 4) using Hydra v9.5 from the Ubuntu 22.04 attacker VM (192.168.1.7) against the Wazuh Server SSH service at 192.168.1.6.
- Created a custom password wordlist with 9 entries; Hydra completed the attack in approximately 7 seconds — 8 failed attempts followed by 1 correct password found.
- Wazuh detected all 8 failed SSH login attempts in real time from /var/log/auth.log, generating individual Rule 5760 (Level 5) alerts for each failed attempt.
- Wazuh automatically correlated the 8 failed attempts from source IP 192.168.1.7 and escalated to Rule 5763 at Level 10 — a Brute Force detection alert mapped to MITRE ATT&CK T1110 (Brute Force) and the Credential Access tactic.
- Authentication success alert (Rule 5715) was generated when Hydra successfully authenticated using the correct password.
- Source attacker IP (192.168.1.7) was captured and recorded in all alert metadata, demonstrating Wazuh's ability to attribute attacks to a specific source.
- Compliance frameworks triggered by the brute force detection: GDPR, HIPAA, PCI-DSS, and NIST 800-53.
- Installed Medusa v2.2 on the attacker VM as an additional brute force tool for future simulations.

Issues Faced & Resolutions – Week 12

- **1. Confirming SSH service was accessible and accepting connections from the attacker VM**

  Issue: Before running Hydra, it was necessary to confirm that the SSH service on the Wazuh Server (192.168.1.6) was running and reachable from the attacker VM (192.168.1.7) on the same internal network.

  Resolution: Ran a manual SSH connection test from the attacker VM to 192.168.1.6 to confirm the port was open and responding. Verified that openssh-server was active on the Ubuntu Server using 'sudo systemctl status ssh'. Proceeded with the Hydra simulation once connectivity was confirmed.

- **2. Ensuring the custom wordlist included the correct target password**

Issue: The custom wordlist needed to include a known password that would succeed so that the brute force simulation could demonstrate both failed attempts and a successful authentication event (Rule 5715). An incorrect wordlist would only show failed attempts.

Resolution: Created the wordlist manually, placing the correct password as the last entry so Hydra would go through all failed attempts before finding it. This produced the expected detection sequence in Wazuh: multiple Rule 5760 alerts followed by Rule 5763 escalation and Rule 5715 success.

- **3. Verifying that Wazuh was reading /var/log/auth.log for SSH event detection**

  Issue: Before starting the attack, it was important to confirm that Wazuh's log analysis module was actively monitoring /var/log/auth.log on the Ubuntu Server — the source of all SSH authentication events. Without this, Hydra's activity would not appear in the Wazuh dashboard.

  Resolution: Checked the Wazuh manager's ossec.conf to confirm the localfile configuration included /var/log/auth.log with the syslog log format. Verified by checking recent auth.log entries after the Hydra run and confirming all events appeared in the Wazuh Security Events dashboard.

Tools Used – Week 12

- **Hydra v9.5 (Ubuntu 22.04 Attacker VM) –** Primary brute force tool used to simulate the SSH credential attack against the Wazuh Server. Run with a custom password wordlist targeting port 22.
- **Medusa v2.2 (Ubuntu 22.04 Attacker VM) –** Alternative brute force tool installed this week for use in future attack simulations.
- **Wazuh SIEM (Manager + Indexer + Dashboard v4.7.5) –** Primary detection platform used to monitor /var/log/auth.log, generate per-attempt alerts (Rule 5760), correlate into brute force detection (Rule 5763), and capture the successful login (Rule 5715).
- **Ubuntu Server 24.04 (VM, 192.168.1.6) –** Target host; runs the Wazuh SIEM stack and the SSH service that was attacked during the simulation.
- **Ubuntu 22.04 (Attacker VM, 192.168.1.7) –** Source of the brute force attack; Hydra and Medusa were run from this machine.
- **VirtualBox –** Hypervisor hosting both the Ubuntu Server and attacker VM on an internal network.

Conclusion of Week 12

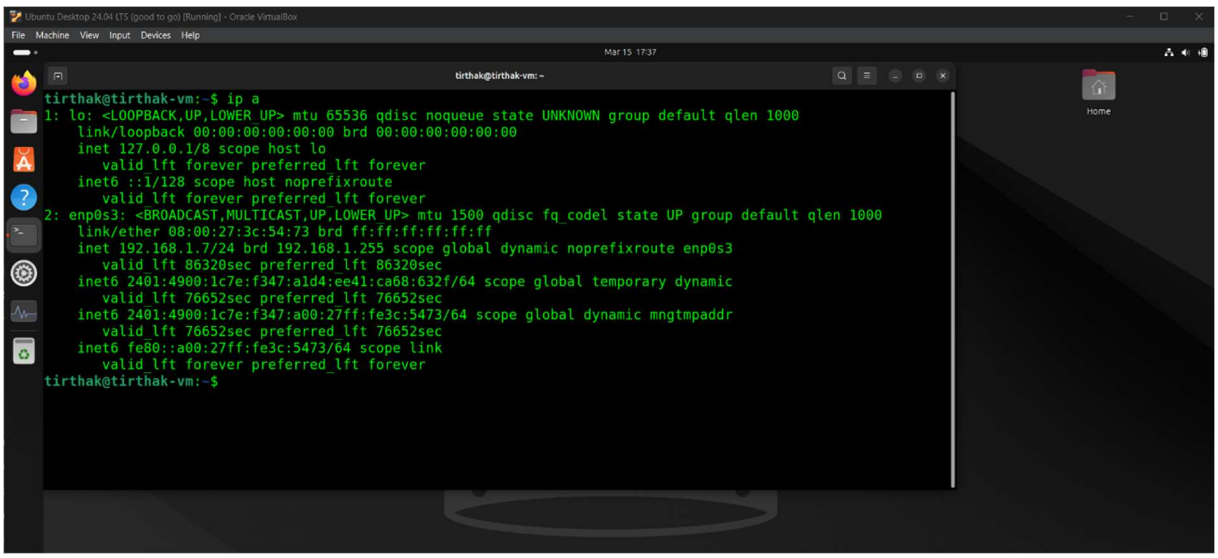
Week 12 successfully completed the fourth attack simulation of the internship lab project. Using Hydra v9.5 from the Ubuntu 22.04 attacker VM, an SSH brute force attack was carried out against the Wazuh Server, simulating a real-world credential attack. Wazuh performed exactly as expected — generating individual failed login alerts (Rule 5760, Level 5), automatically escalating to a correlated Brute Force detection alert (Rule 5763, Level 10) mapped to MITRE T1110, and firing a successful authentication alert (Rule 5715) when the correct password was found. The attacker IP was captured in all alert metadata, demonstrating end-to-end attack attribution. Compliance frameworks including GDPR, HIPAA, PCI-

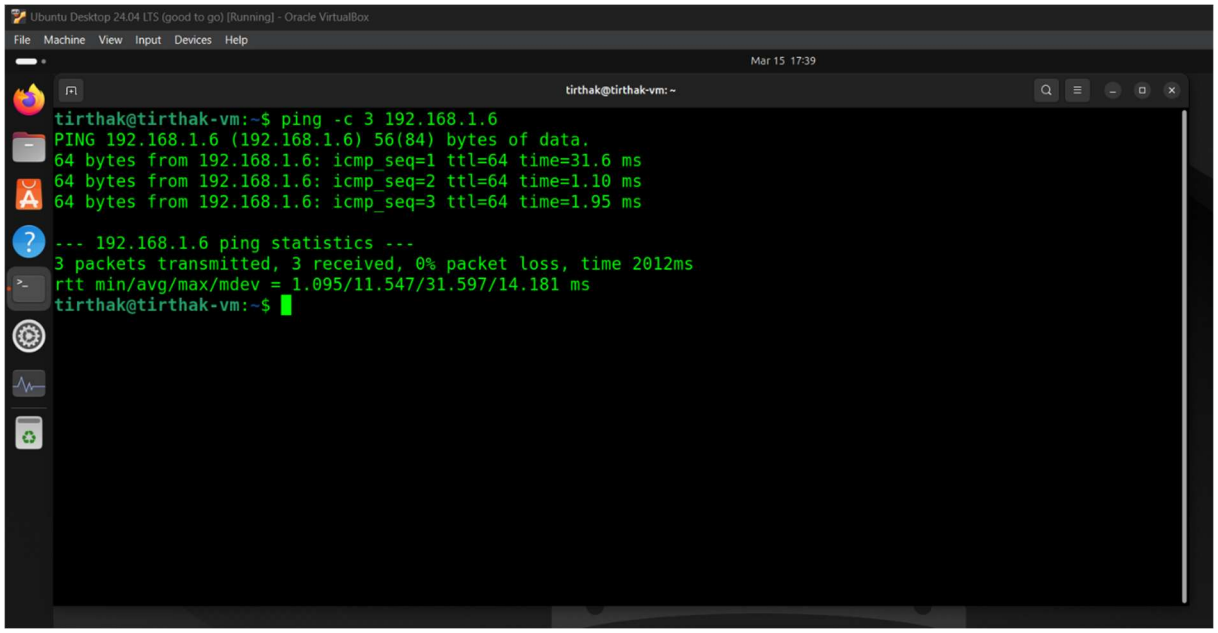
DSS, and NIST 800-53 were triggered by the detection, reinforcing the real-world relevance of the simulation.

With the completion of Week 12, the lab now has four fully documented attack simulations: Failed Windows Login (Event ID 4625, MITRE T1078), EICAR Test File Detection, FIM File Creation Trigger (Rule 554), and SSH Brute Force (MITRE T1110, Rule 5763). Medusa v2.2 was also installed on the attacker VM for use in future simulations. The upcoming weeks will focus on documentation, incident response write-ups, and internship report preparation.

Screenshots of Activities (if Any)

[Please insert screenshots here — Hydra terminal output showing 8 failed and 1 success, Rule 5760 Level 5 alert in Wazuh dashboard, Rule 5763 Level 10 Brute Force alert, Rule 5715 authentication success alert, MITRE T1110 tag in alert metadata, source IP 192.168.1.7 in alert details]





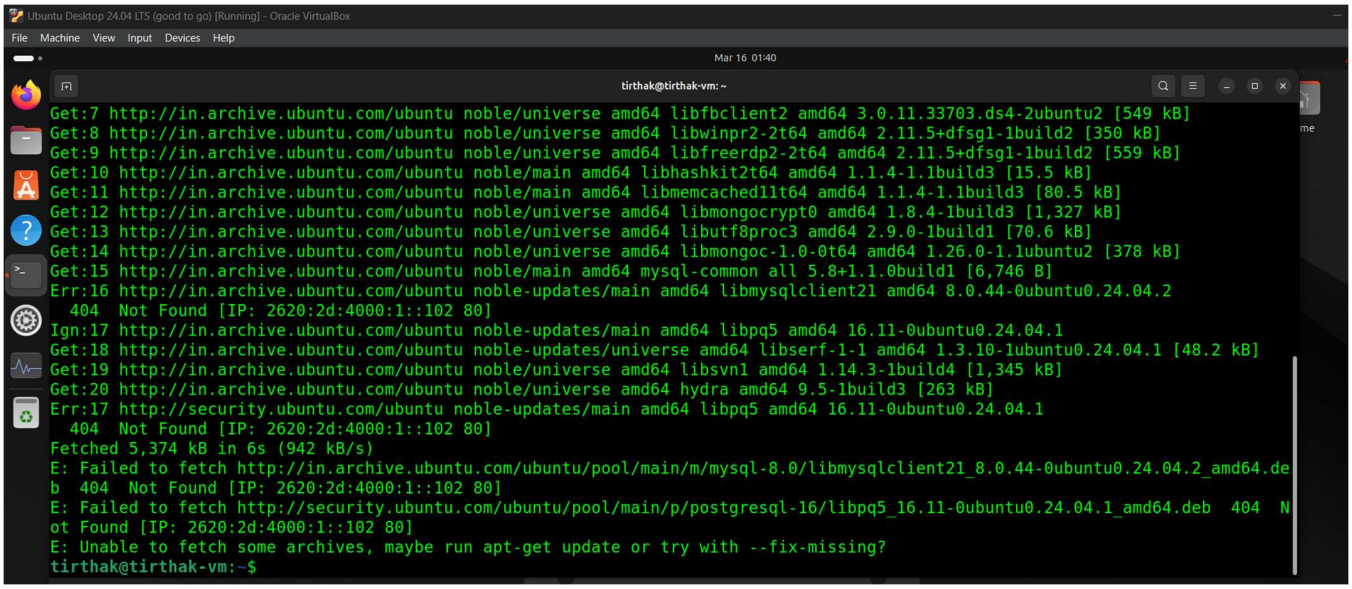


Fig: some packages are giving trouble in installing the Hydra tool

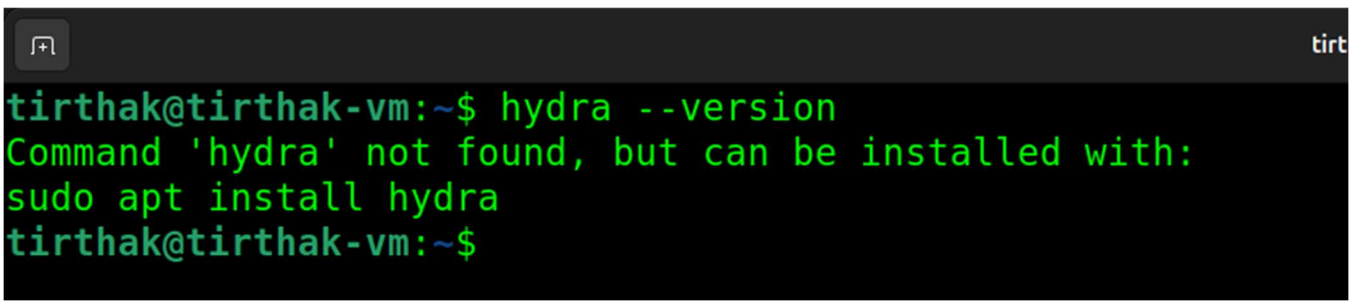
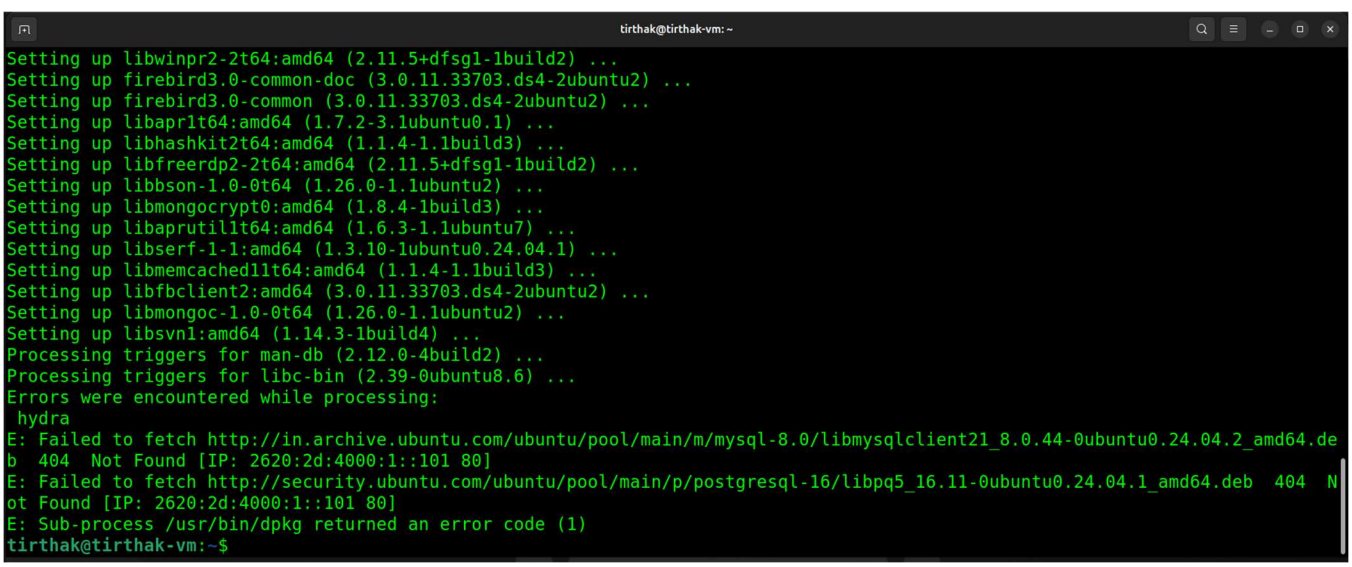


Fig: Hydra is Installed but configuration files miss some of its imp files therefore giving Trouble.



```
tirthak@tirthak-vm: ~
tirthak@tirthak-vm:~$ sudo apt-get install -y hydra --no-install-recommends
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
hydra is already the newest version (9.5-1build3).
You might want to run 'apt --fix-broken install' to correct these.
The following packages have unmet dependencies:
 hydra : Depends: libmysqlclient21 (>= 8.0.11) but it is not going to be installed
         Depends: libpq5 but it is not going to be installed
E: Unmet dependencies. Try 'apt --fix-broken install' with no packages (or specify a solution).
tirthak@tirthak-vm:~$
```

Fig: Trying to install Hydra missing files

```
tirthak@tirthak-vm: ~
Reading state information... Done
Correcting dependencies... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  libmysqlclient21 libpq5
The following NEW packages will be installed:
  libmysqlclient21 libpq5
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
1 not fully installed or removed.
Need to get 1,399 kB of archives.
After this operation, 7,294 kB of additional disk space will be used.
Err:1 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 libmysqlclient21 amd64 8.0.44-0ubuntu0.24.04.2
  404  Not Found [IP: 2620:2d:4000:1::102 80]
Ign:2 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 libpq5 amd64 16.11-0ubuntu0.24.04.1
Err:2 http://security.ubuntu.com/ubuntu noble-updates/main amd64 libpq5 amd64 16.11-0ubuntu0.24.04.1
  404  Not Found [IP: 2620:2d:4000:1::102 80]
E: Failed to fetch http://in.archive.ubuntu.com/ubuntu/pool/main/m/mysql-8.0/libmysqlclient21_8.0.44-0ubuntu0.24.04.2_amd64.de
b  404  Not Found [IP: 2620:2d:4000:1::102 80]
E: Failed to fetch http://security.ubuntu.com/ubuntu/pool/main/p/postgresql-16/libpq5_16.11-0ubuntu0.24.04.1_amd64.deb  404  N
ot Found [IP: 2620:2d:4000:1::102 80]
E: Unable to fetch some archives, maybe run apt-get update or try with --fix-missing?
tirthak@tirthak-vm:~$
```

Fig: We have tried installing Hydra's Missing files but we got some trouble and have to perform some troubleshooting in order to make the installation happen.

```
tirthak@tirthak-vm: ~
0 upgraded, 2 newly installed, 0 to remove and 178 not upgraded.
1 not fully installed or removed.
Need to get 1,400 kB of archives.
After this operation, 7,294 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 libmysqlclient21 amd64 8.0.45-0ubuntu0
.24.04.1 [1,255 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 libpq5 amd64 16.13-0ubuntu0.24.04.1 [1
46 kB]
Fetched 1,400 kB in 2s (696 kB/s)
Selecting previously unselected package libmysqlclient21:amd64.
(Reading database ... 185644 files and directories currently installed.)
Preparing to unpack .../libmysqlclient21_8.0.45-0ubuntu0.24.04.1_amd64.deb ...
Unpacking libmysqlclient21:amd64 (8.0.45-0ubuntu0.24.04.1) ...
Selecting previously unselected package libpq5:amd64.
Preparing to unpack .../libpq5_16.13-0ubuntu0.24.04.1_amd64.deb ...
Unpacking libpq5:amd64 (16.13-0ubuntu0.24.04.1) ...
Setting up libmysqlclient21:amd64 (8.0.45-0ubuntu0.24.04.1) ...
Setting up libpq5:amd64 (16.13-0ubuntu0.24.04.1) ...
Setting up hydra (9.5-1build3) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
tirthak@tirthak-vm:~$
```

Fig:We have successfully installed and trouble shooted Hydra v9.5 in our Ubuntu 24.04 Attacker machine.

```
tirthak@tirthak-vm: ~
tirthak@tirthak-vm:~$ hydra -version
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service
organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

[ERROR] unknown mode i for option -e, only supporting "n", "s" and "r"
tirthak@tirthak-vm:~$
tirthak@tirthak-vm:~$
tirthak@tirthak-vm:~$ medusa -version
Medusa v2.2 [http://www.foofus.net] (C) JoMo-Kun / Foofus Networks <jmk@foofus.net>

ALERT: Host information must be supplied.
tirthak@tirthak-vm:~$
```

Fig: As you can se we have successfully installed Hydra and you can see its version as well and we have also installed Medusa a similar tool as hydra for backup just inn case hydra gave us some trouble in the future.

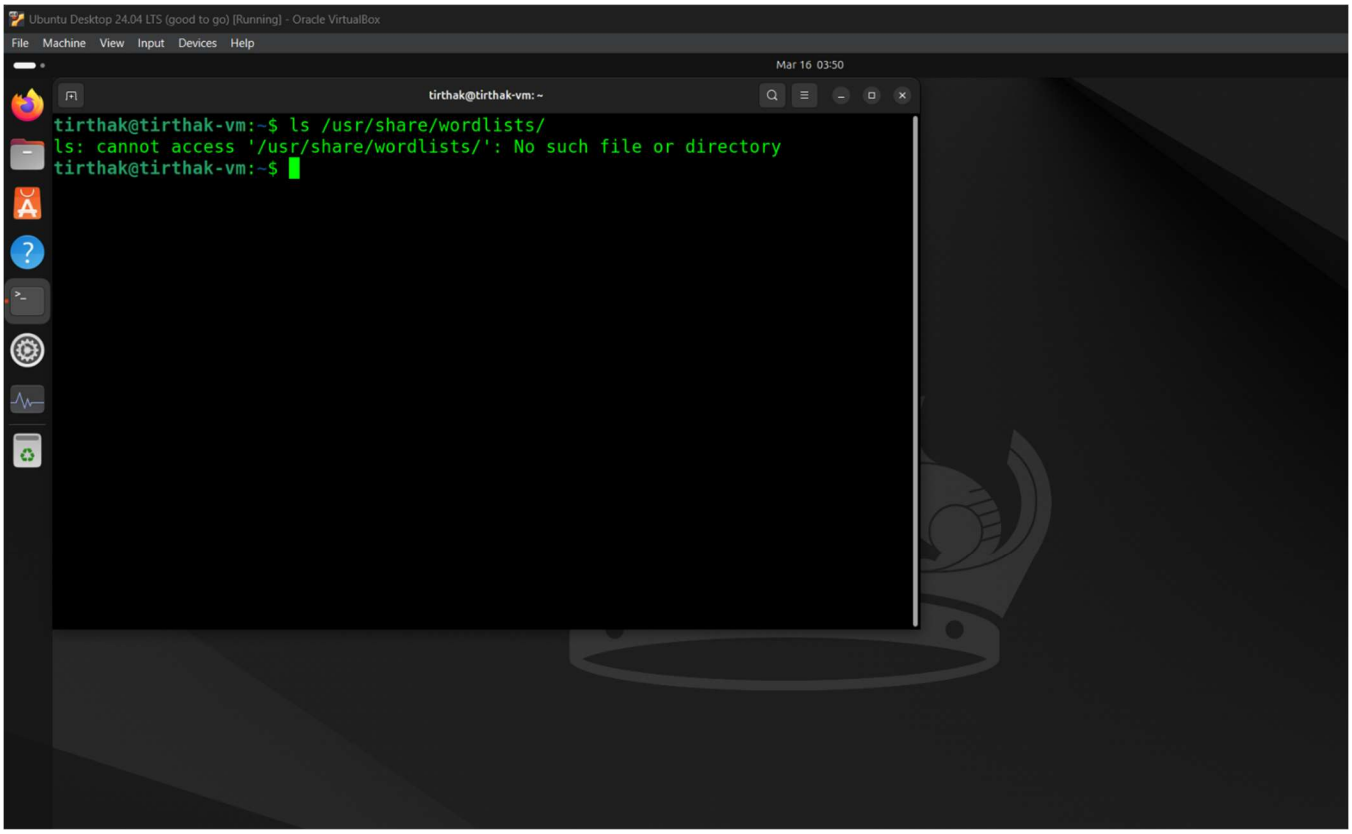


Fig: - we are checking for any word list if any already available in our attacker machine which we can use for Brute force rapid password attempts but we haven't found any so we will make our custom one.

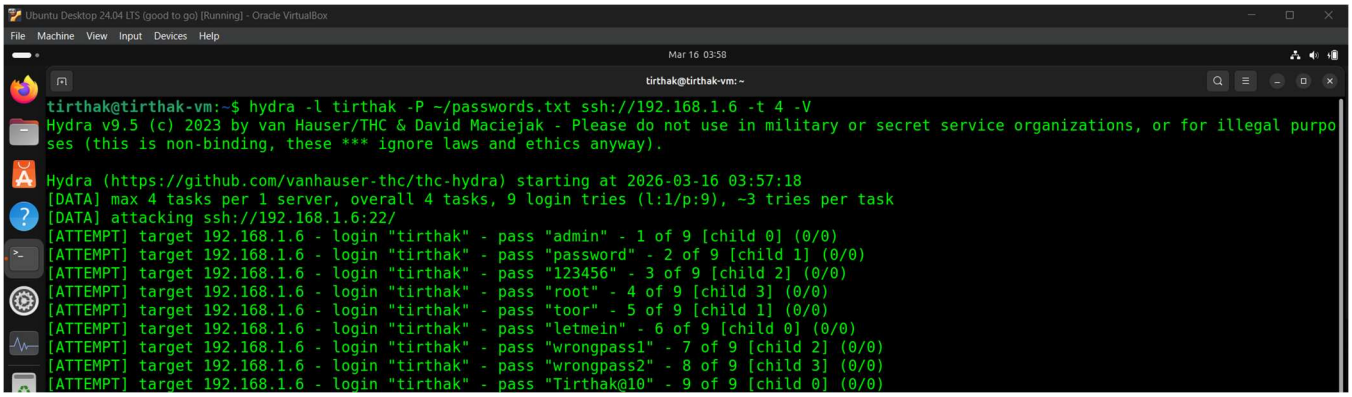
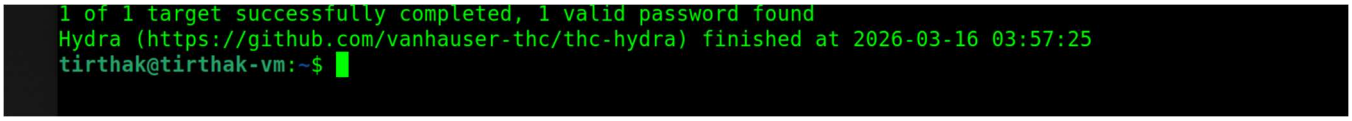


Fig: - We have made our custom password list which we will be using for brute forcing rapid password attempts on our Wazuh SIEM which is deployed on our Ubuntu Server 22.04 so if we are able to get the access of Ubuntu server, we get the access of Wazuh SIEM and can compromise it.



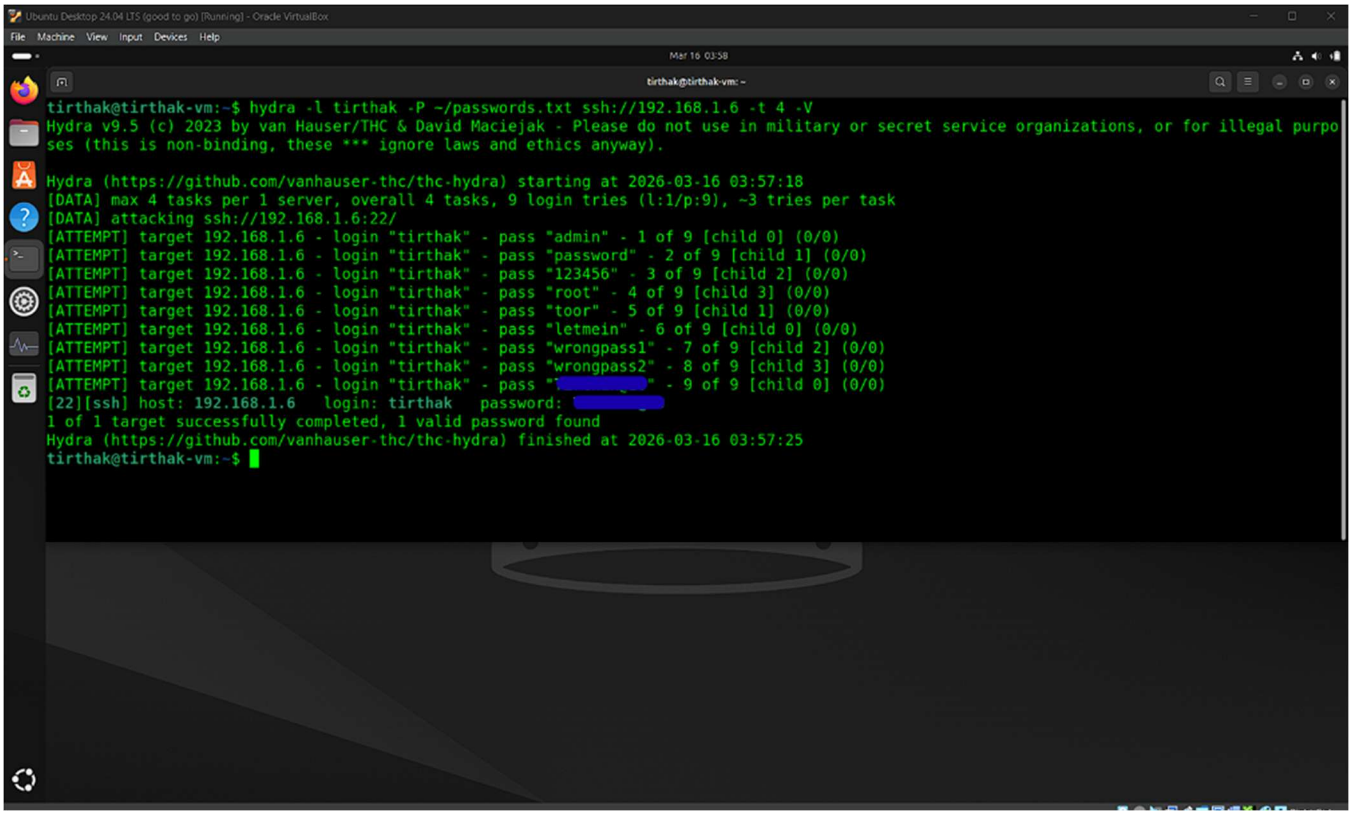
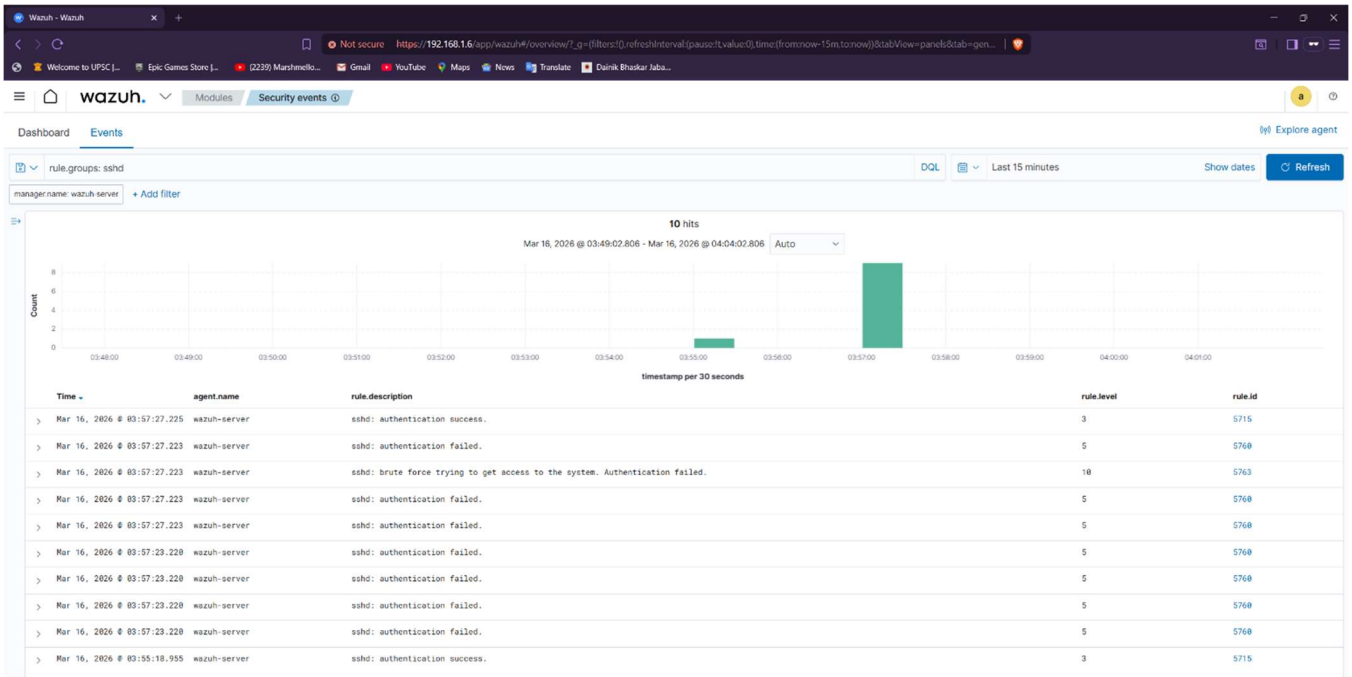
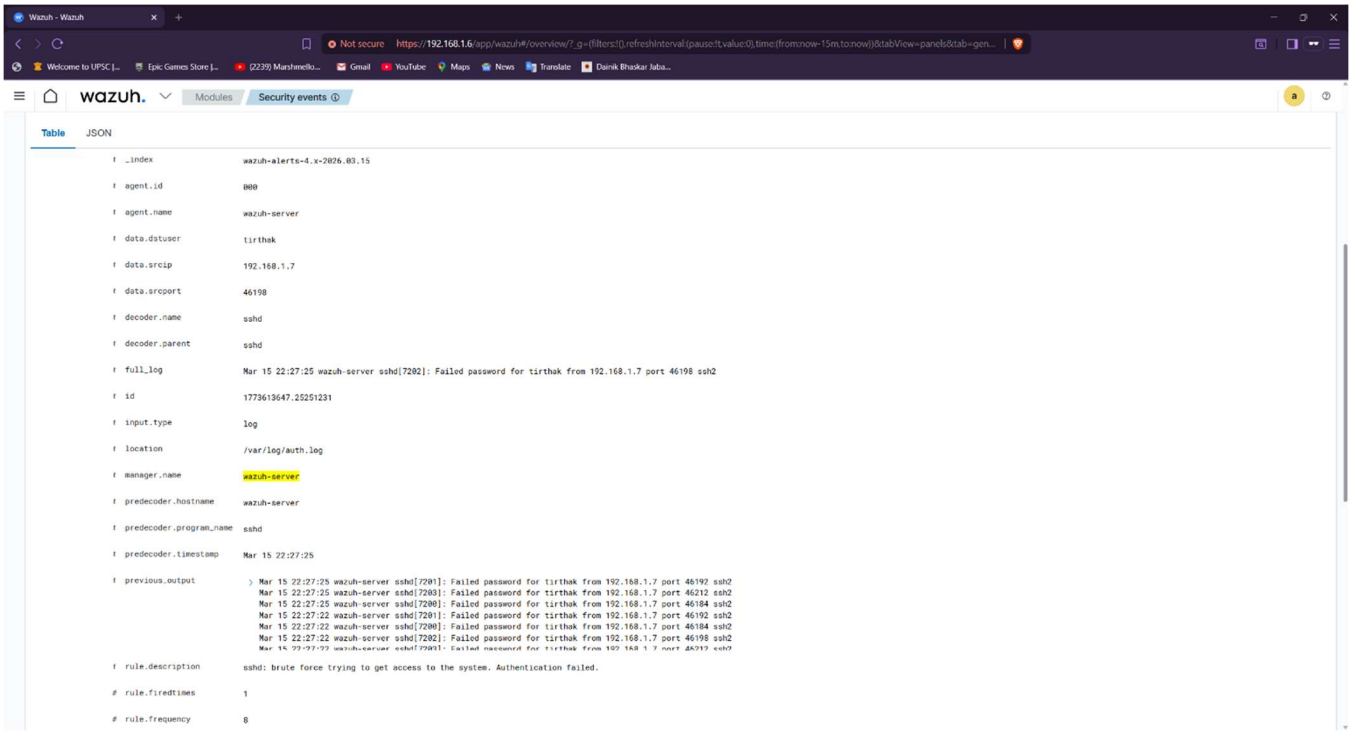
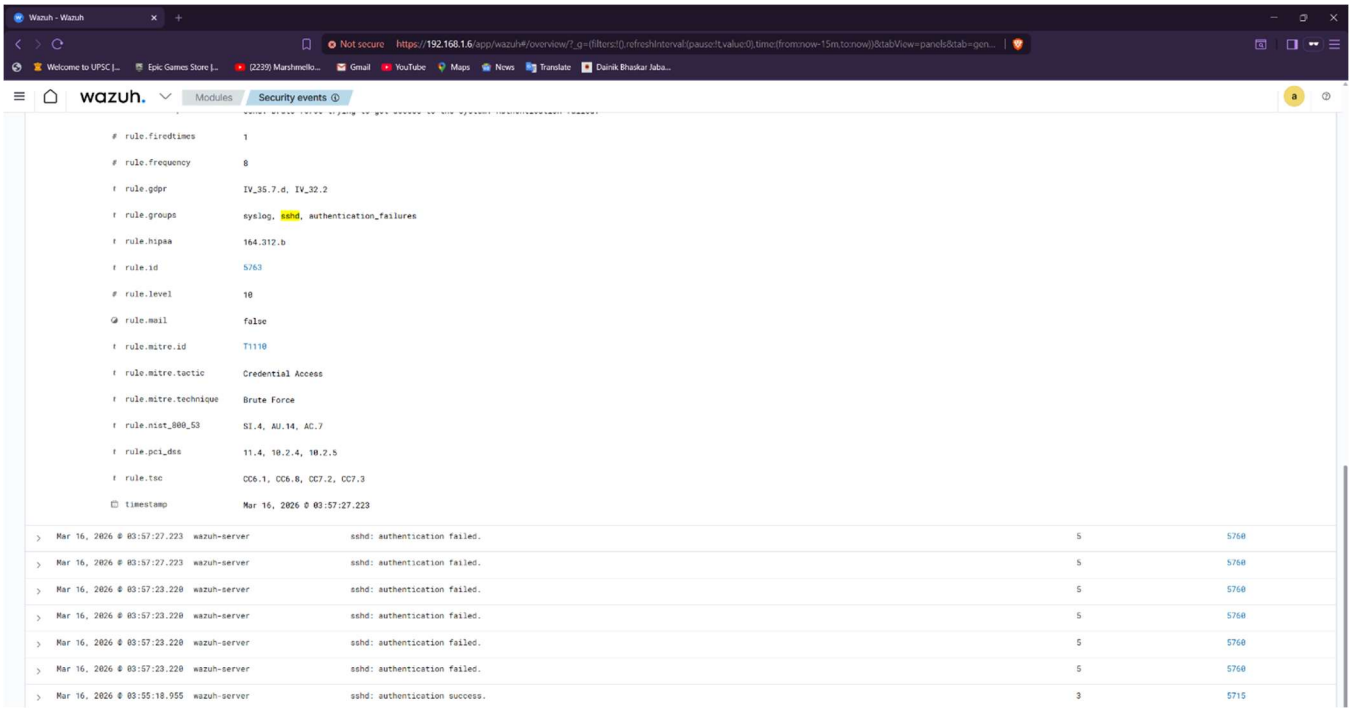


Fig: - As you can se we have tried our custom password list and u can see we were able to successfully compromise our Ubuntu Server.



Fig;- As u can see we have got the hit of SSH Brute force and Authentication failed insode our Wazuh SIEM.


Wazuh - Wazuh
Not secure https://192.168.1.6/app/wazuh#/overview/?_g=(filters:!(),refreshInterval:(pause:!t,value:0),time:(from:now-15m,to:now))&tabView=panels&tab=gen...
wazuh.
Modules
Security events
Table
JSON
_index wazuh-alerts-4.x-2026.03.15
agent.id 000
agent.name wazuh-server
data.dstuser tirthak
data.srcip 192.168.1.7
data.srcport 46198
decoder.name sshd
decoder.parent sshd
full_log Mar 15 22:27:25 wazuh-server sshd[7202]: Failed password for tirthak from 192.168.1.7 port 46198 ssh2
id 1773613647.25251231
input.type log
location /var/log/auth.log
manager.name wazuh-server
predecoder.hostname wazuh-server
predecoder.program_name sshd
predecoder.timestamp Mar 15 22:27:25
previous_output Mar 15 22:27:25 wazuh-server sshd[7201]: Failed password for tirthak from 192.168.1.7 port 46192 ssh2
Mar 15 22:27:25 wazuh-server sshd[7203]: Failed password for tirthak from 192.168.1.7 port 46212 ssh2
Mar 15 22:27:25 wazuh-server sshd[7200]: Failed password for tirthak from 192.168.1.7 port 46184 ssh2
Mar 15 22:27:22 wazuh-server sshd[7201]: Failed password for tirthak from 192.168.1.7 port 46192 ssh2
Mar 15 22:27:22 wazuh-server sshd[7200]: Failed password for tirthak from 192.168.1.7 port 46184 ssh2
Mar 15 22:27:22 wazuh-server sshd[7202]: Failed password for tirthak from 192.168.1.7 port 46198 ssh2
rule.description sshd: brute force trying to get access to the system. Authentication failed.
rule.firedtimes 1
rule.frequency 8


Wazuh - Wazuh
Not secure https://192.168.1.6/app/wazuh#/overview/?_g=(filters:!(),refreshInterval:(pause:!t,value:0),time:(from:now-15m,to:now))&tabView=panels&tab=gen...
wazuh.
Modules
Security events
rule.firedtimes 1
rule.frequency 8
rule.gdpr IV_35.7.d, IV_32.2
rule.groups syslog, sshd, authentication_failures
rule.hipaa 164.312.b
rule.id 5763
rule.level 10
rule.mail false
rule.mitre.id T1110
rule.mitre.tactic Credential Access
rule.mitre.technique Brute Force
rule.nist_800_53 SI.4, AU.14, AC.7
rule.pci_dss 11.4, 10.2.4, 10.2.5
rule.tsc CC6.1, CC6.8, CC7.2, CC7.3
timestamp Mar 16, 2026 @ 03:57:27.223
Mar 16, 2026 @ 03:57:27.223 wazuh-server sshd: authentication failed. 5 5760
Mar 16, 2026 @ 03:57:27.223 wazuh-server sshd: authentication failed. 5 5760
Mar 16, 2026 @ 03:57:23.220 wazuh-server sshd: authentication failed. 5 5760
Mar 16, 2026 @ 03:57:23.220 wazuh-server sshd: authentication failed. 5 5760
Mar 16, 2026 @ 03:57:23.220 wazuh-server sshd: authentication failed. 5 5760
Mar 16, 2026 @ 03:57:23.220 wazuh-server sshd: authentication failed. 5 5760
Mar 16, 2026 @ 03:55:18.955 wazuh-server sshd: authentication success. 3 5715

Alerts generated:

Rule ID	Description	Level
5760	sshd: authentication failed	5
5763	sshd: brute force trying to get access	**10**
5715	sshd: authentication success	3

Rule 5763 Full Alert Details:

Field	Value
Rule ID	5763
Rule Level	10 (High)
Description	sshd: brute force trying to get access to the system
Source IP	192.168.1.7 (attacker VM)
Target User	tirthak
Log Source	/var/log/auth.log
Rule Frequency	8 attempts triggered the alert
MITRE ID	T1110
MITRE Technique	Brute Force
MITRE Tactic	Credential Access

Compliance tags:

Framework	Tags
GDPR	IV_35.7.d, IV_32.2
HIPAA	164.312.b
PCI-DSS	11.4, 10.2.4, 10.2.5
NIST 800-53	SI.4, AU.14, AC.7

What made it special:

- Wazuh didn't just detect individual failures
- It correlated 8 failed attempts from 4 different ports (46184, 46192, 46198, 46212) into ONE brute force alert
- That's real SIEM correlation — exactly how enterprise SOCs work